

Physical Chemistry (I)

Lecture 18

Phase Diagrams of Solutions



In the Last Class

- Solution models
 - Ideal solutions
 - Ideal-dilute solutions
 - Regular solutions
 - Bragg-Williams model
- Deviations from ideality
 - Excess function
 - Activity
 - Activity of BW model



Last class

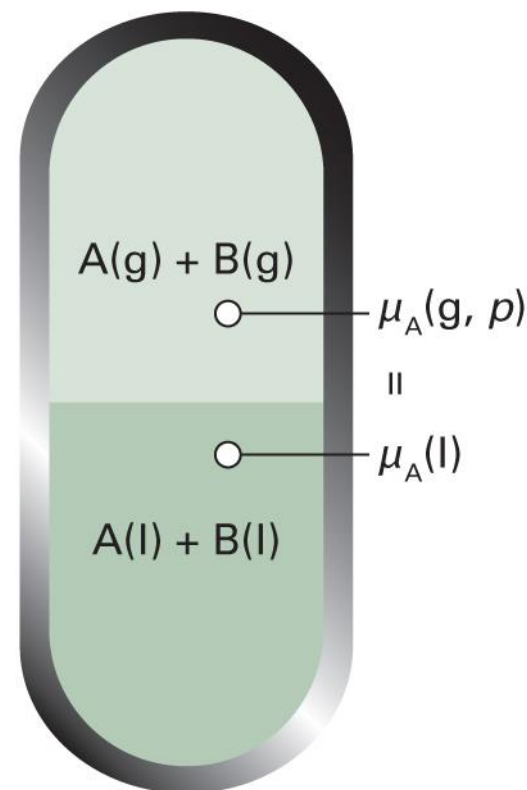
Mixing of Liquids

$$\mu_A(l) = \mu_A(g)$$

Using the chemical potential of perfect gas,

$$\mu_A(l) = \mu_A^*(l) + RT \ln(p_A/p_A^*)$$

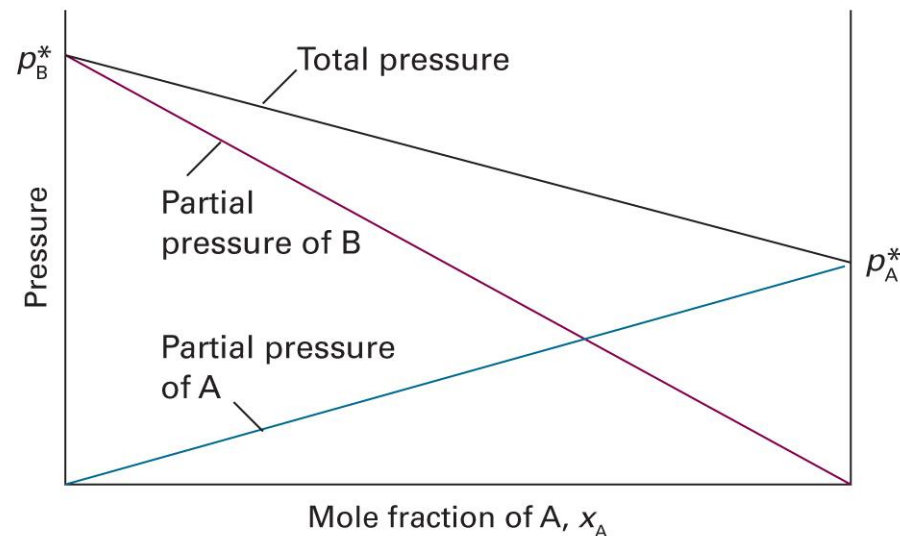
(General relationship between μ_A and p_A)



Raoult's Law: Ideal Solution

$$p_A = x_A p_A^*$$

- **Ideal solutions:** mixtures that obey the law throughout the composition range from pure A to pure B



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5A.11)

Mixing of Ideal Solution

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

(Same with perfect gas mixtures)

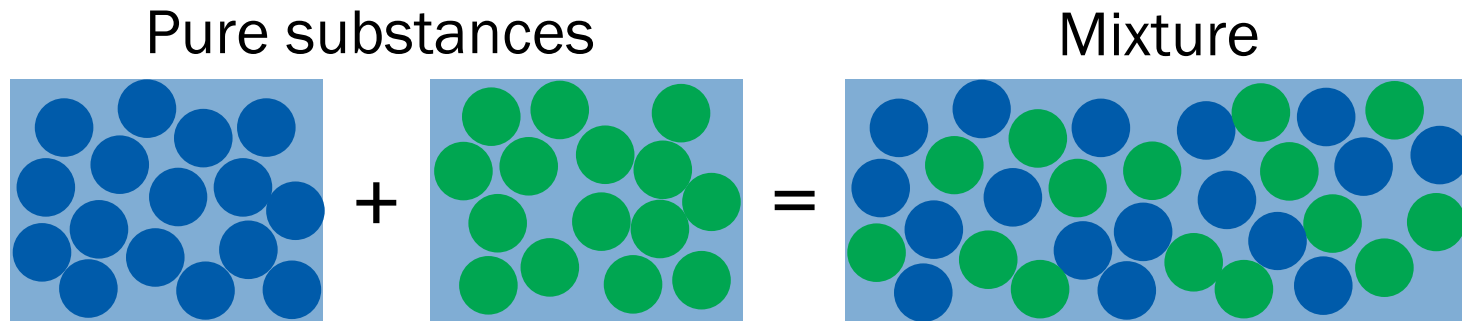
Last class

Henry's Law

$$p_B = x_B K_B$$

- **Ideal-dilute solutions:** mixtures for which the solute B obeys Henry's law and the solvent A obeys Raoult's law

Regular Solution



A model solution with

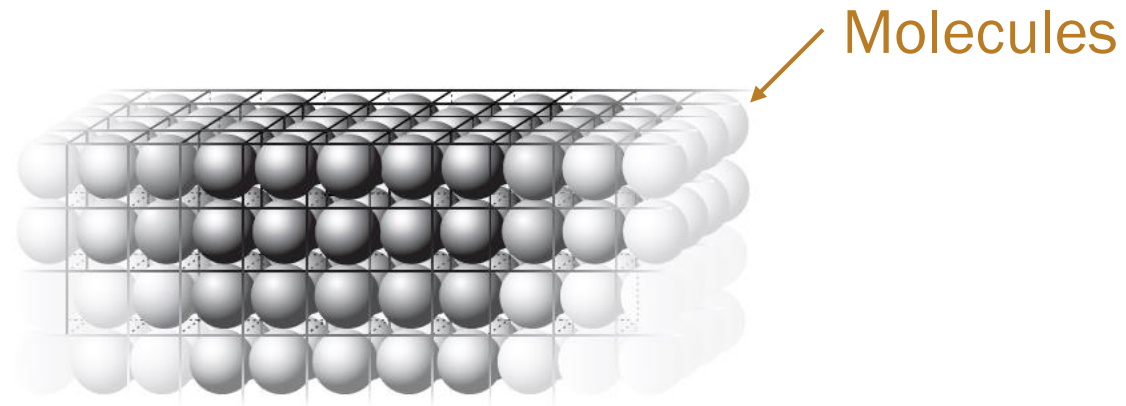
$$\Delta_{\text{mix}}S = \Delta_{\text{mix}}S^{\text{ideal}} = -nR(x_A \ln x_A + x_B \ln x_B)$$

and

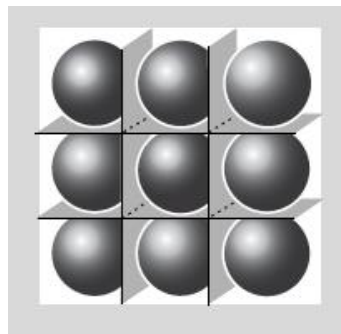
$$\Delta_{\text{mix}}H \neq 0$$

Last class

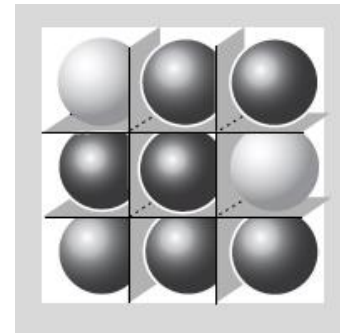
Bragg-Williams Model



Pure substance



Mixture



(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figures 14.1 and 16.2)

Last class

Bragg-Williams Model

$$\Delta_{\text{mix}}S = -nR(x_A \ln x_A + x_B \ln x_B)$$

$$\Delta_{\text{mix}}U = \xi nRT x_A x_B$$

$$\Delta_{\text{mix}}A = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$

Deviations from Ideality

- Excess function

$$\chi^E = \Delta_{\text{mix}}X - \Delta_{\text{mix}}X^{\text{ideal}}$$

- Activity

$$\mu_A = \mu_A^* + RT \ln a_A$$

Activity a_J

- Ideal-dilute solution

- Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln x_A$

- Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln x_B$

- where $\mu_B^\ominus(l) = \mu_B^*(l) + RT \ln(K_B/p_B^*)$

- Real solution

- Solvent: $\mu_A(l) = \mu_A^*(l) + RT \ln a_A$

- Solute: $\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B$

Activity Coefficient

$$\gamma_J = a_J/x_J$$

- Solvent

$$\mu_A(l) = \mu_A^*(l) + RT \ln a_A = \mu_A^*(l) + RT \ln x_A + RT \ln \gamma_A$$

- Solute

$$\mu_B(l) = \mu_B^\ominus(l) + RT \ln a_B = \mu_B^\ominus(l) + RT \ln x_B + RT \ln \gamma_B$$

As $x_A \rightarrow 1$ and $x_B \rightarrow 0$, $a_J \rightarrow x_J$ and $\gamma_J \rightarrow 1$

Activity of BW Model

$$x_A \ln \gamma_A + x_B \ln \gamma_B = \xi x_A x_B$$

The following equations satisfy the given criterion:

$$\ln \gamma_A = \xi x_B^2, \text{ or } \gamma_A = e^{\xi x_B^2}$$

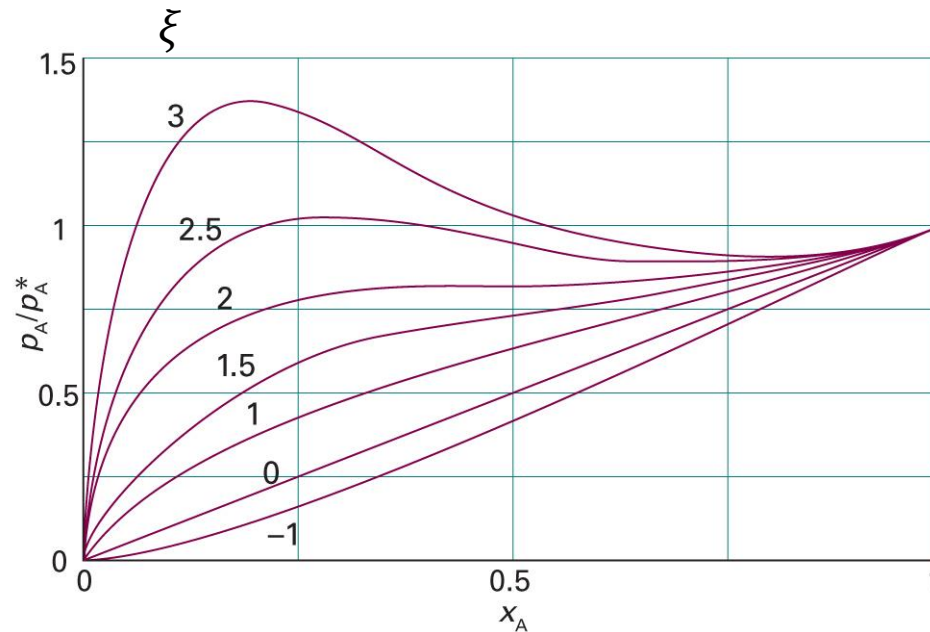
$$\ln \gamma_B = \xi x_A^2, \text{ or } \gamma_B = e^{\xi x_A^2}$$

(Margules equations)

Last class

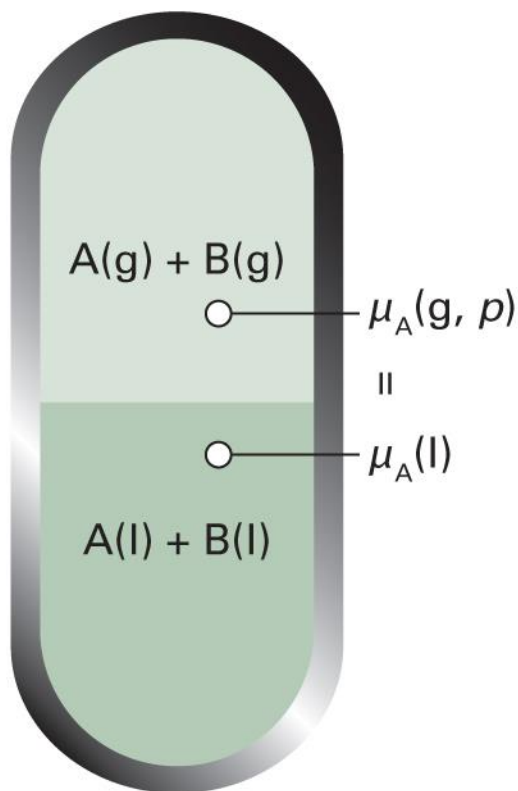
Vapor Pressure of BW Model

$$p_A/p_A^* = x_A e^{\xi(1-x_A)^2}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5F.2)

Liquid-Vapor Equilibrium



- Constraints

$$\mu_A(g) = \mu_A(l)$$

$$\mu_B(g) = \mu_B(l)$$

- May have $x_A(g) \neq x_A(l)$

Brief Illustration 5A.3

Consider a mixture of benzene and methylbenzene, which form an approximately ideal solution. The vapour pressure of pure benzene at 20°C is 75.0 Torr and that of pure methylbenzene is 21.0 Torr at the same temperature. What are the mole fractions in the vapour when the two liquids are in an equimolar mixture?

In an equimolar mixture,

$$x_{\text{benzene}}(l) = x_{\text{methylbenzene}}(l) = 0.5$$

Thus,

$$p_{\text{benzene}} = 75.0 \text{ Torr} \times 0.5 = 37.5 \text{ Torr}$$

$$p_{\text{methylbenzene}} = 21.0 \text{ Torr} \times 0.5 = 10.5 \text{ Torr}$$

Brief Illustration 5A.3

The total pressure is $37.5 \text{ Torr} + 10.5 \text{ Torr} = 48.0 \text{ Torr}$.

From Dalton's law,

$$x_{\text{benzene}}(g) = (37.5 \text{ Torr}) / (48.0 \text{ Torr}) = 0.78$$

$$x_{\text{methylbenzene}}(g) = (10.5 \text{ Torr}) / (48.0 \text{ Torr}) = 0.22$$

Vapor Pressures

From Raoult's law,

$$p_A = x_A p_A^*, \quad p_B = x_B p_B^*$$

The total vapor pressure is

$$p = p_A + p_B = x_A p_A^* + x_B p_B^* = x_A p_A^* + (1 - x_A) p_B^*$$

which becomes

$$p = p_B^* + (p_A^* - p_B^*) x_A$$

Vapor Compositions

$$p = p_B^* + (p_A^* - p_B^*)x_A$$

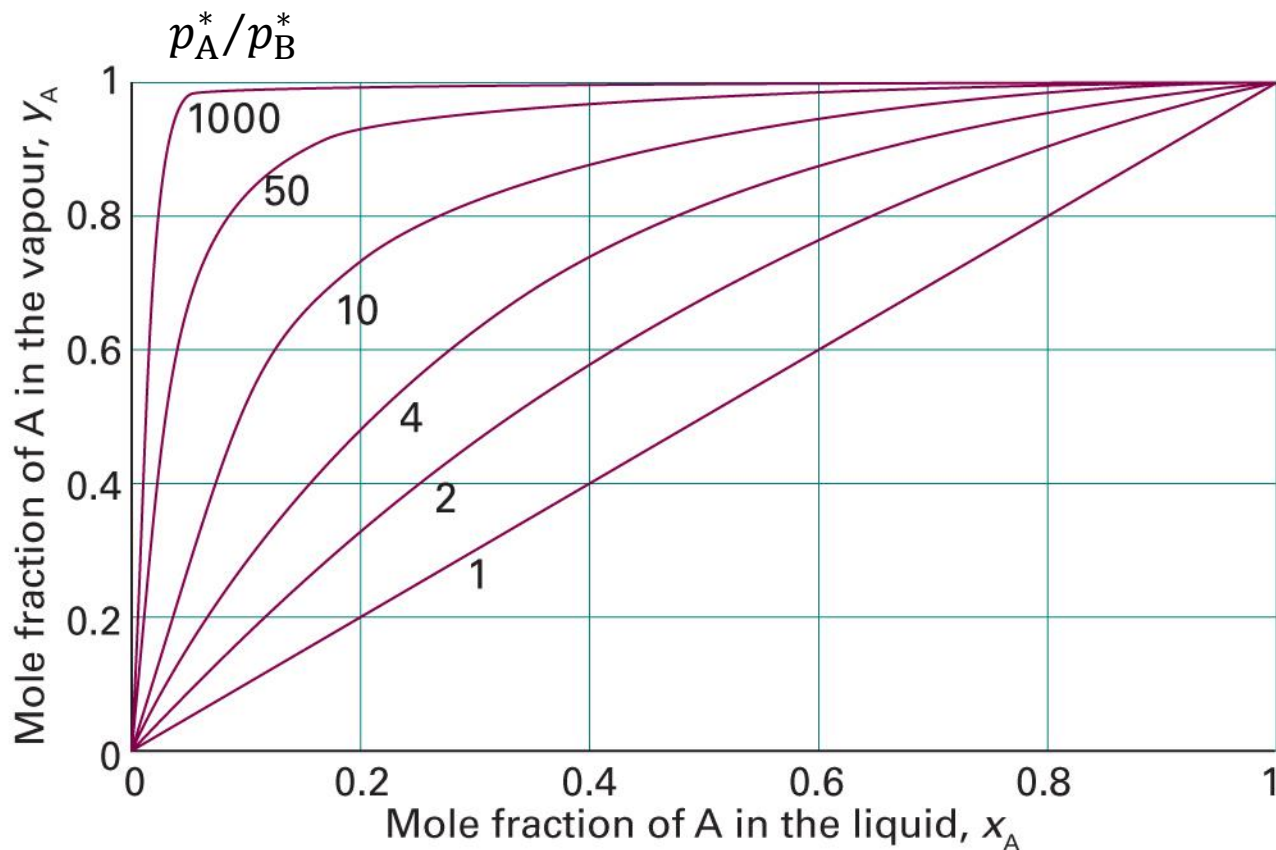
Denote the mole fractions in the vapor as y_A and y_B :

$$y_A = p_A/p, \quad y_B = p_B/p$$

Now,

$$y_A = \frac{x_A p_A^*}{p_B^* + (p_A^* - p_B^*)x_A}, \quad y_B = 1 - y_A$$

Vapor Compositions



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5C.2)

x_A as a Function of y_A

$$y_A = \frac{x_A p_A^*}{p_B^* + (p_A^* - p_B^*)x_A}$$

$$p_B^* y_A + (p_A^* - p_B^*) x_A y_A = x_A p_A^*$$

$$p_B^* y_A = x_A \{p_A^* - (p_A^* - p_B^*) y_A\}$$

Hence,

$$x_A = \frac{y_A p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

Total Vapor Pressure

$$x_A = \frac{y_A p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

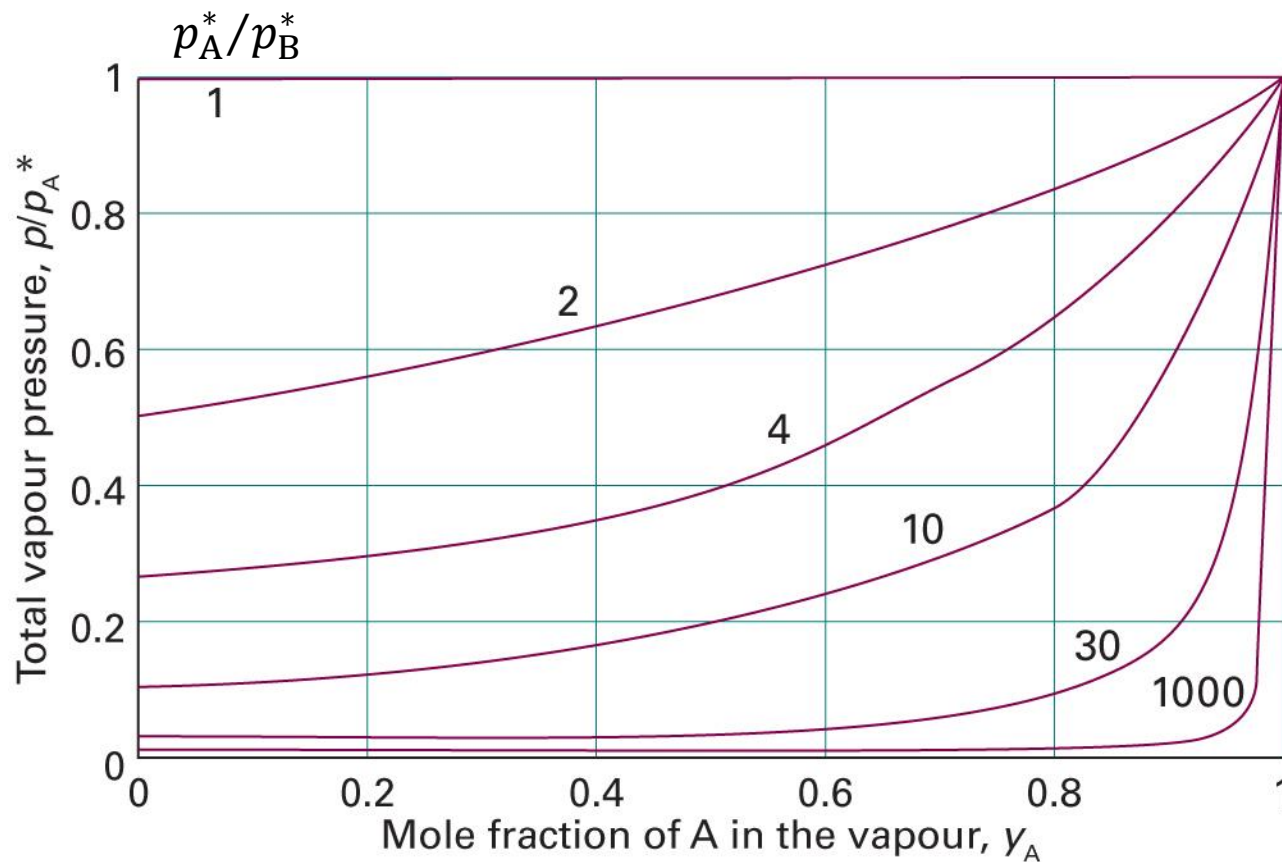
The total vapor pressure is

$$p = p_B^* + (p_A^* - p_B^*) x_A = p_B^* + \frac{(p_A^* - p_B^*) y_A p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

which leads to

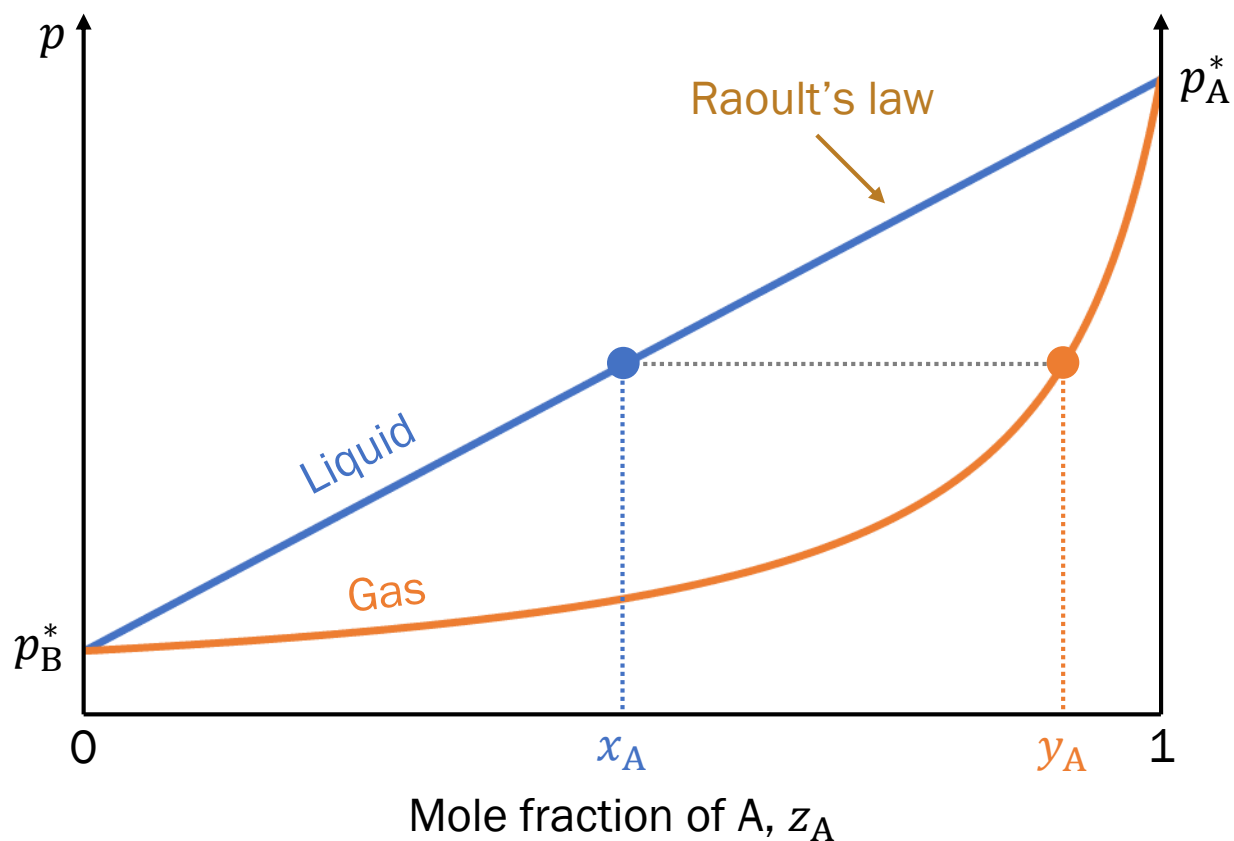
$$p = \frac{p_A^* p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

Total Vapor Pressure



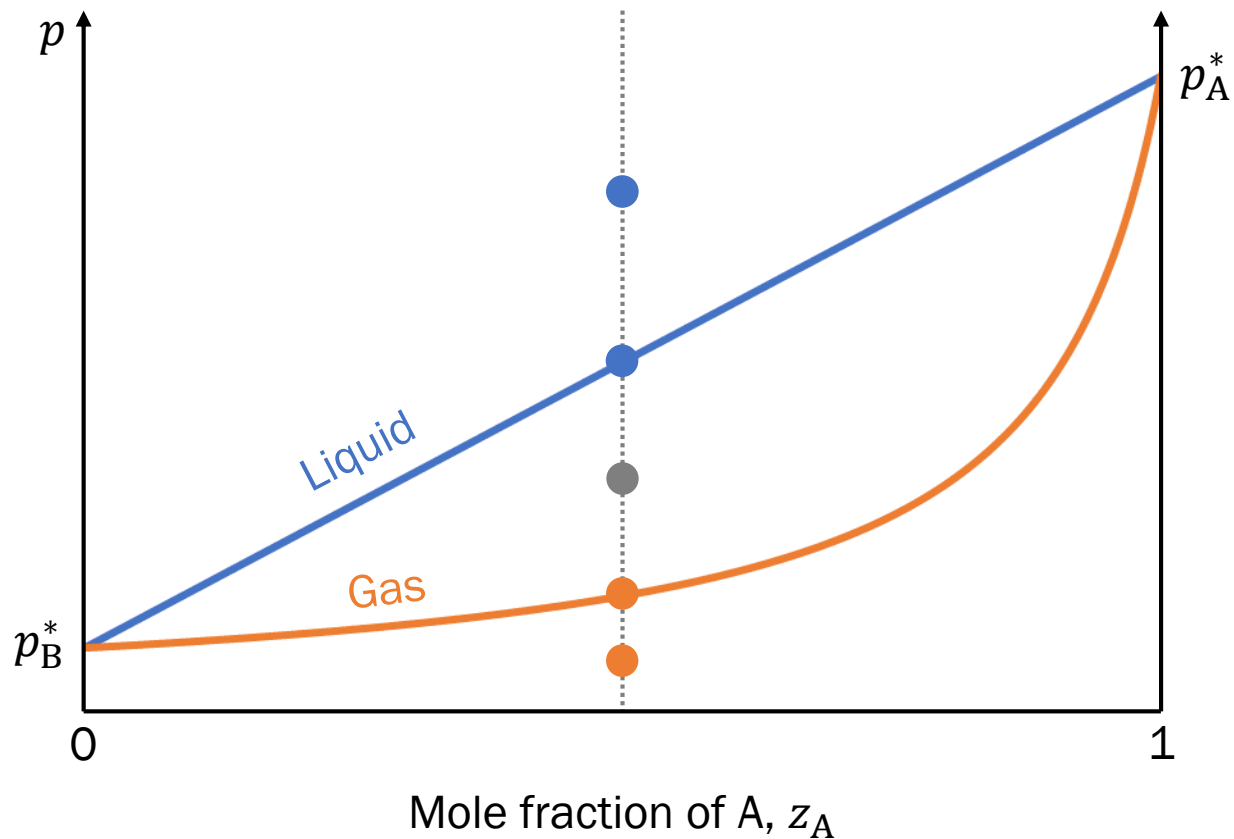
(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 5C.3)

Vapor Pressure Curves

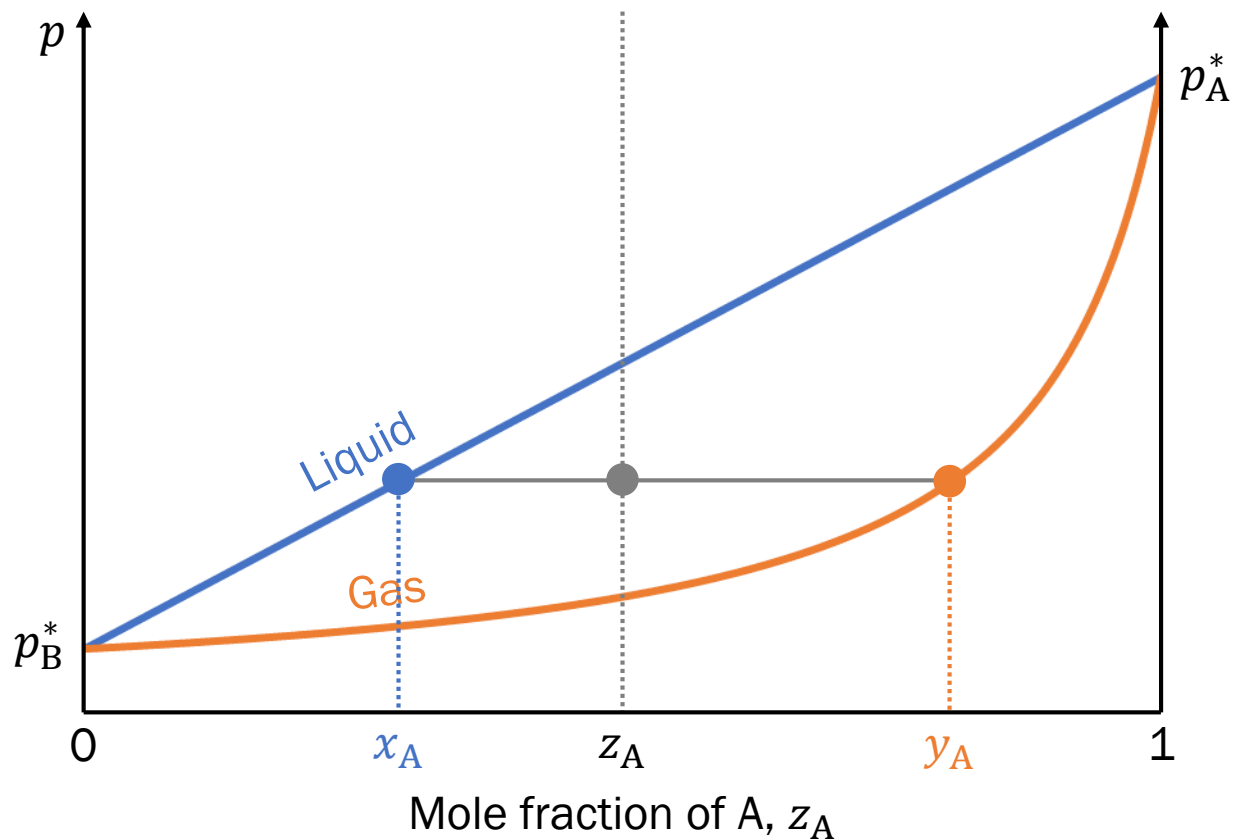


Not in Atkins

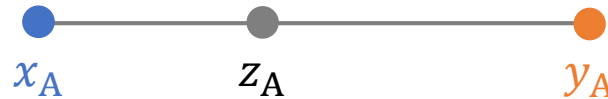
Phase Diagram?



Coexistence: Tie Line

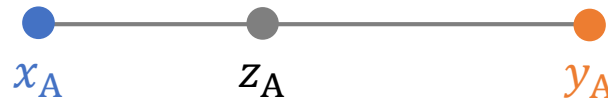


Conservation Laws



- Total amount of A: $n_A = n_{A,l} + n_{A,g}$
- Total amount of B: $n_B = n_{B,l} + n_{B,g}$
- Total mole fraction: $z_A = n_A / (n_A + n_B)$
- Mole fraction at the liquid curve: $x_A = n_{A,l} / (n_{A,l} + n_{B,l})$
- Mole fraction at the gas curve: $y_A = n_{A,g} / (n_{A,g} + n_{B,g})$

Lever Rule



$$z_A = \frac{n_A}{n_A + n_B} = \frac{n_{A,l} + n_{A,g}}{n_{A,l} + n_{A,g} + n_{B,l} + n_{B,g}}$$

$$\begin{aligned} z_A(n_{A,l} + n_{A,g} + n_{B,l} + n_{B,g}) &= n_{A,l} + n_{A,g} \\ &= x_A(n_{A,l} + n_{B,l}) + y_A(n_{A,g} + n_{B,g}) \end{aligned}$$

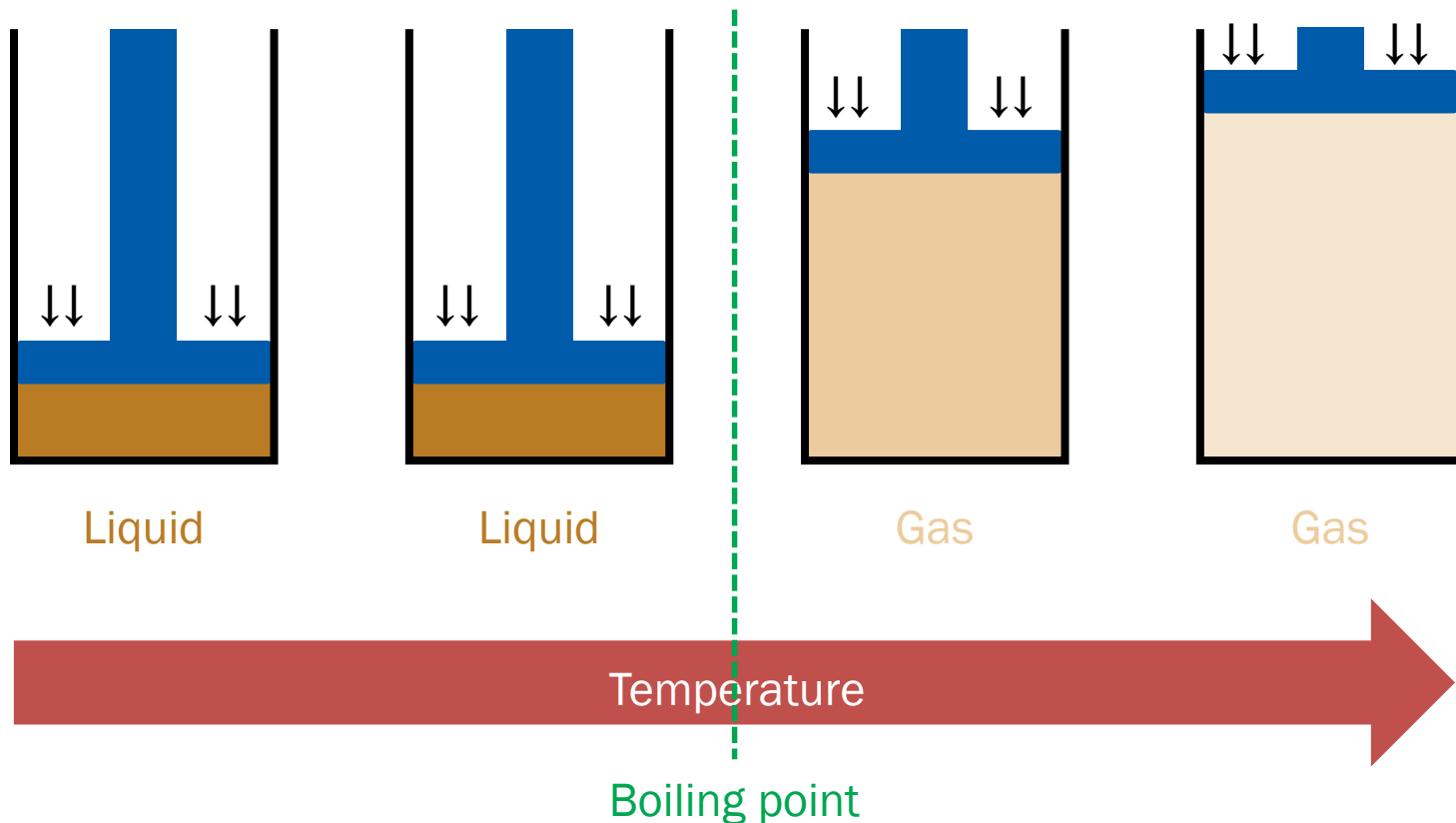
$$(z_A - x_A)(n_{A,l} + n_{B,l}) = (y_A - z_A)(n_{A,g} + n_{B,g})$$

$$(n_{A,g} + n_{B,g}) : (n_{A,l} + n_{B,l}) = (z_A - x_A) : (y_A - z_A)$$

Not in Atkins

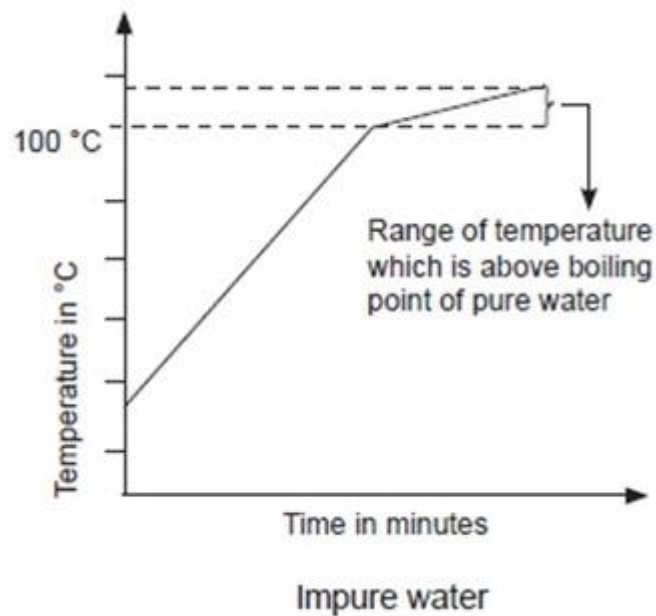
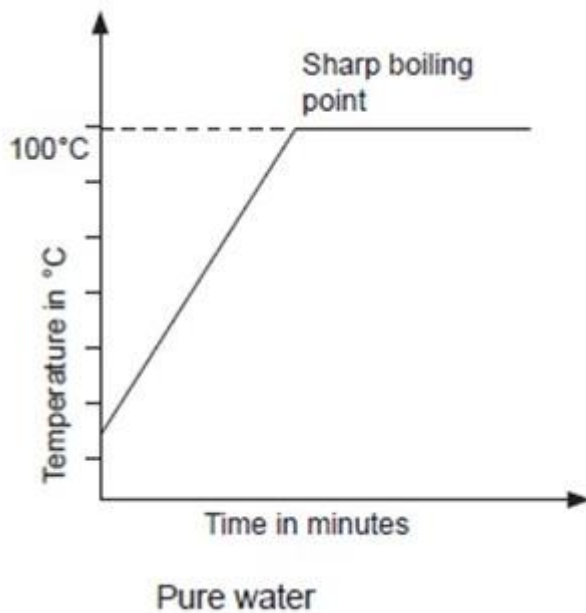
Boiling of Pure Substance

(Isobaric change)



Not in Atkins

Pure vs. Impure



Temperature Dependence

- Compositions depend on p_A^* and p_B^*

$$x_A = \frac{p - p_B^*}{p_B^* - p_A^*}$$

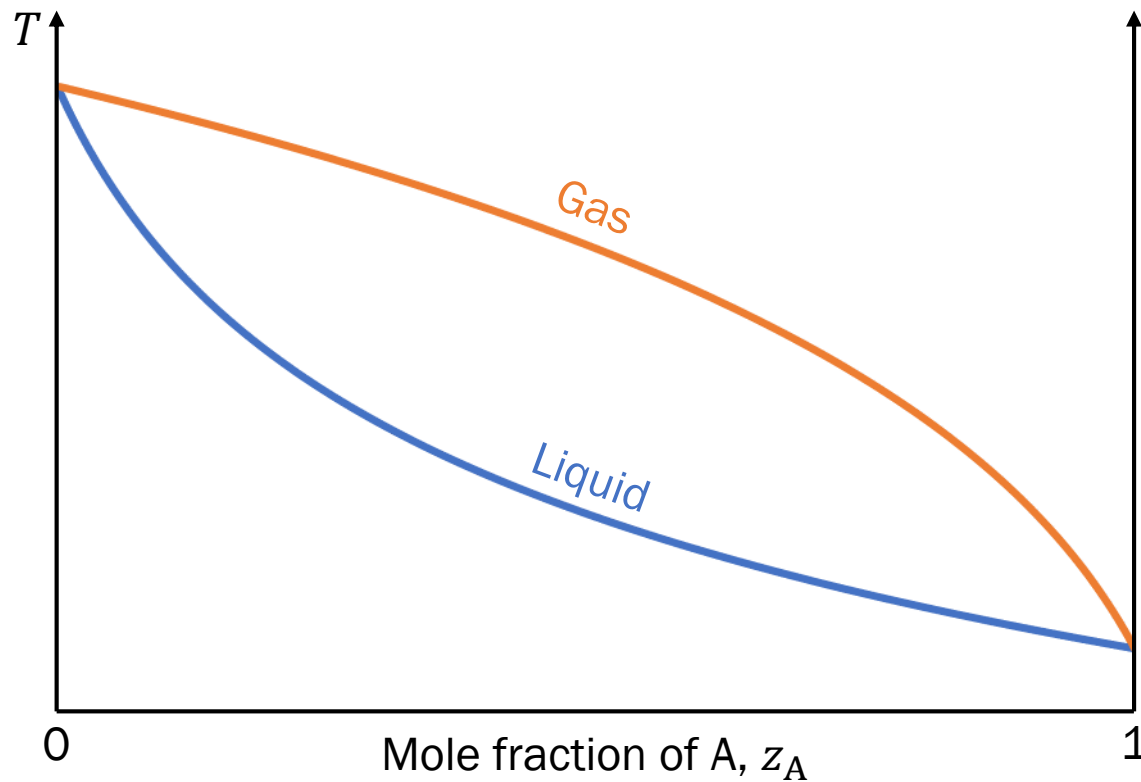
$$y_A = \frac{x_A p_A^*}{p_B^* + (p_A^* - p_B^*) x_A}$$

- Antoine equation: $p_A^* = p_A^*(T)$ and $p_B^* = p_B^*(T)$

$$\log p_J^* = A_J - \frac{B_J}{C_J + T}$$

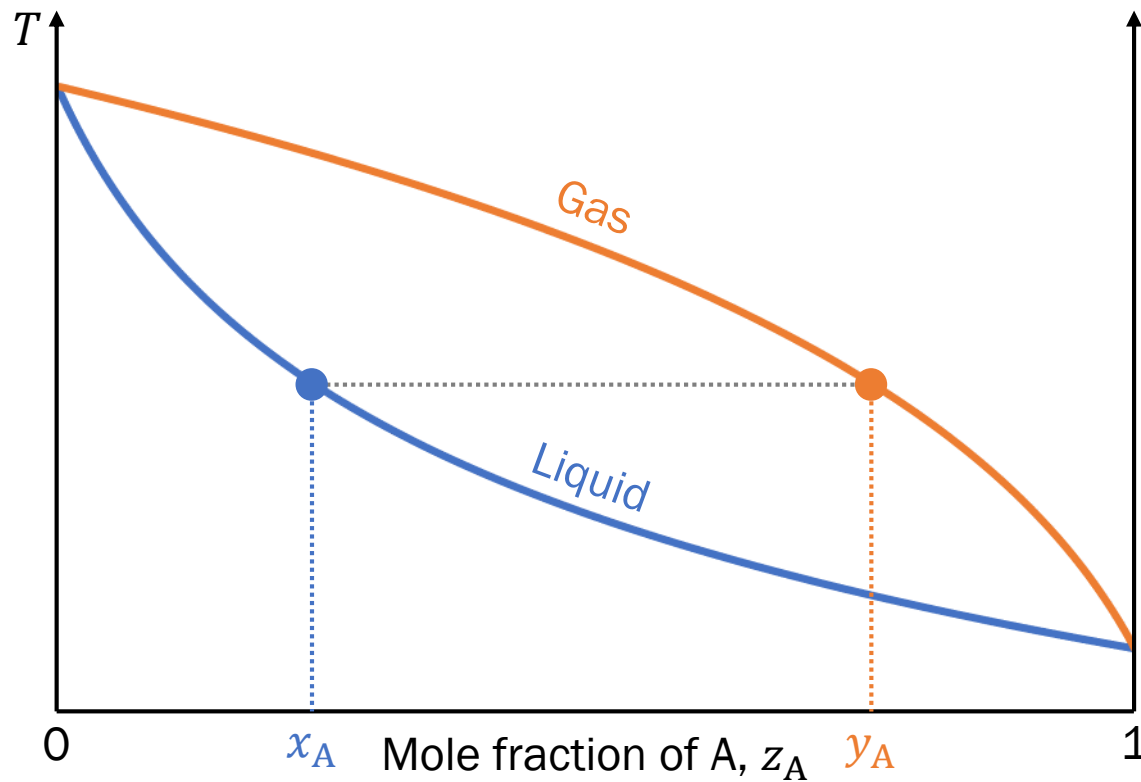
Temperature-Composition

$$p_A^* > p_B^* \text{ (A is more volatile)}$$



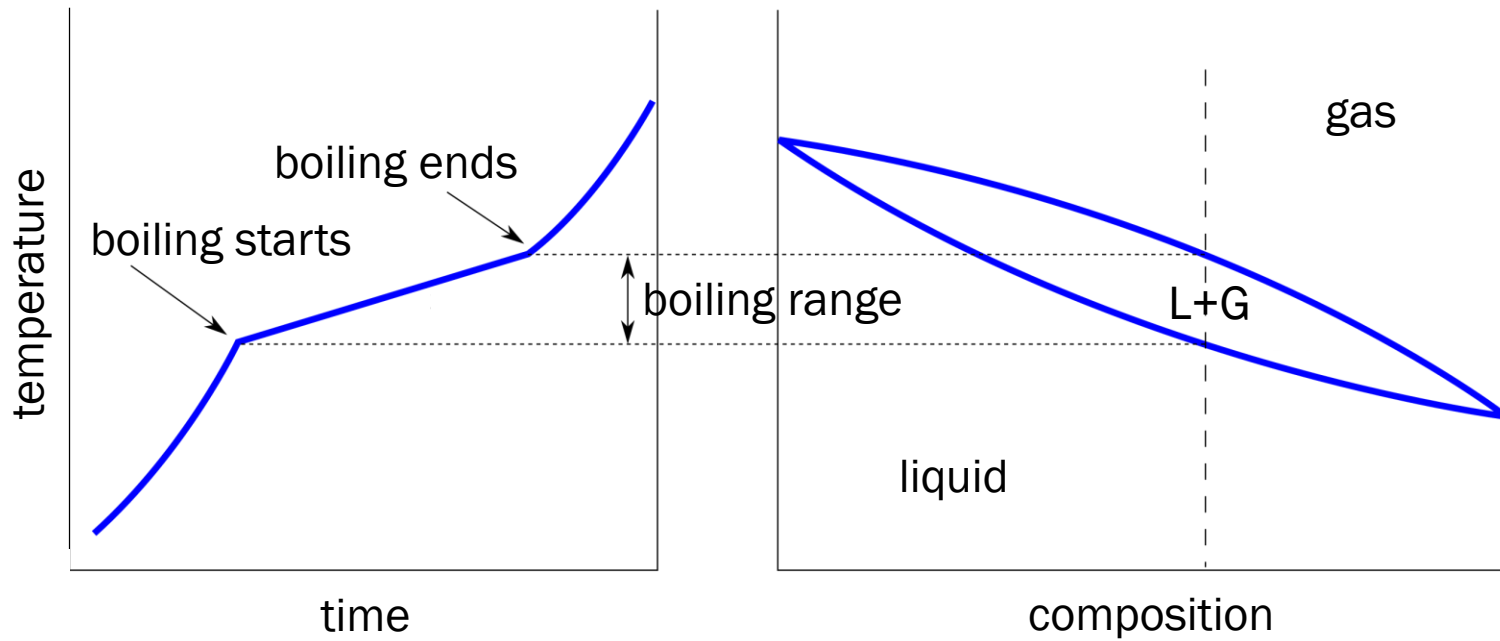
Boiling of Mixture

$p_A^* > p_B^*$ (A is more volatile)



Not in Atkins

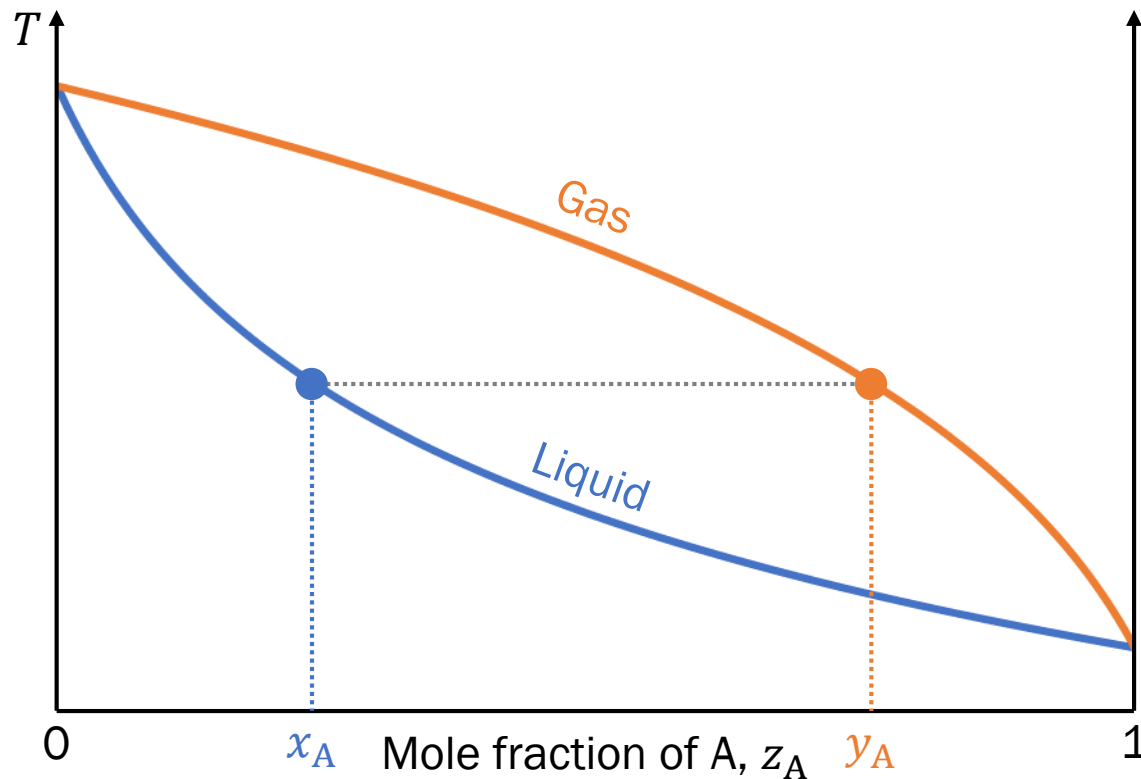
Boiling Curve



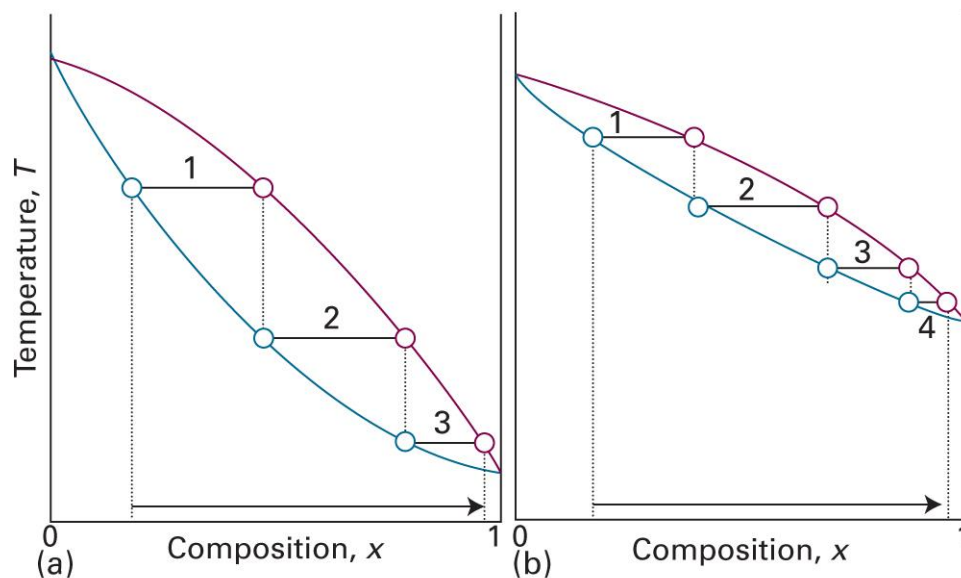
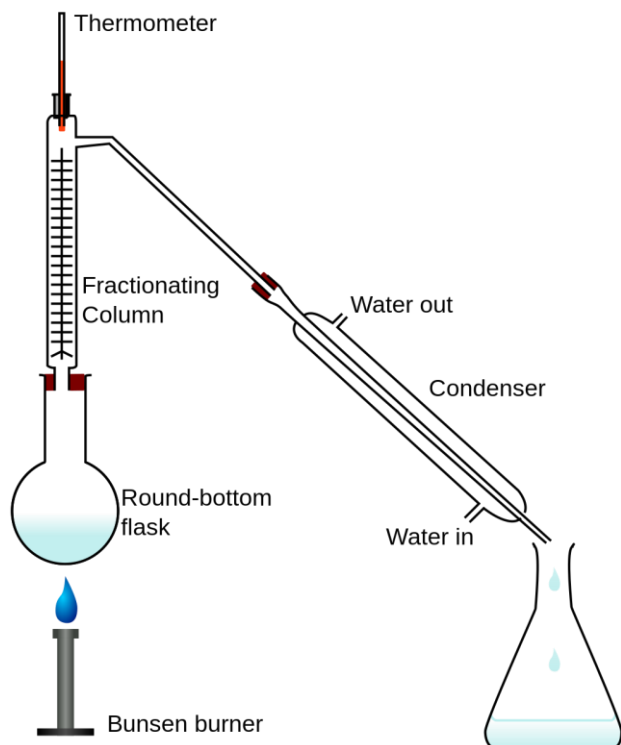
(Image: https://commons.wikimedia.org/wiki/File:Cooling_curve_alloy.svg)

Distillation

- Simple distillation: vapor withdrawn and condensed

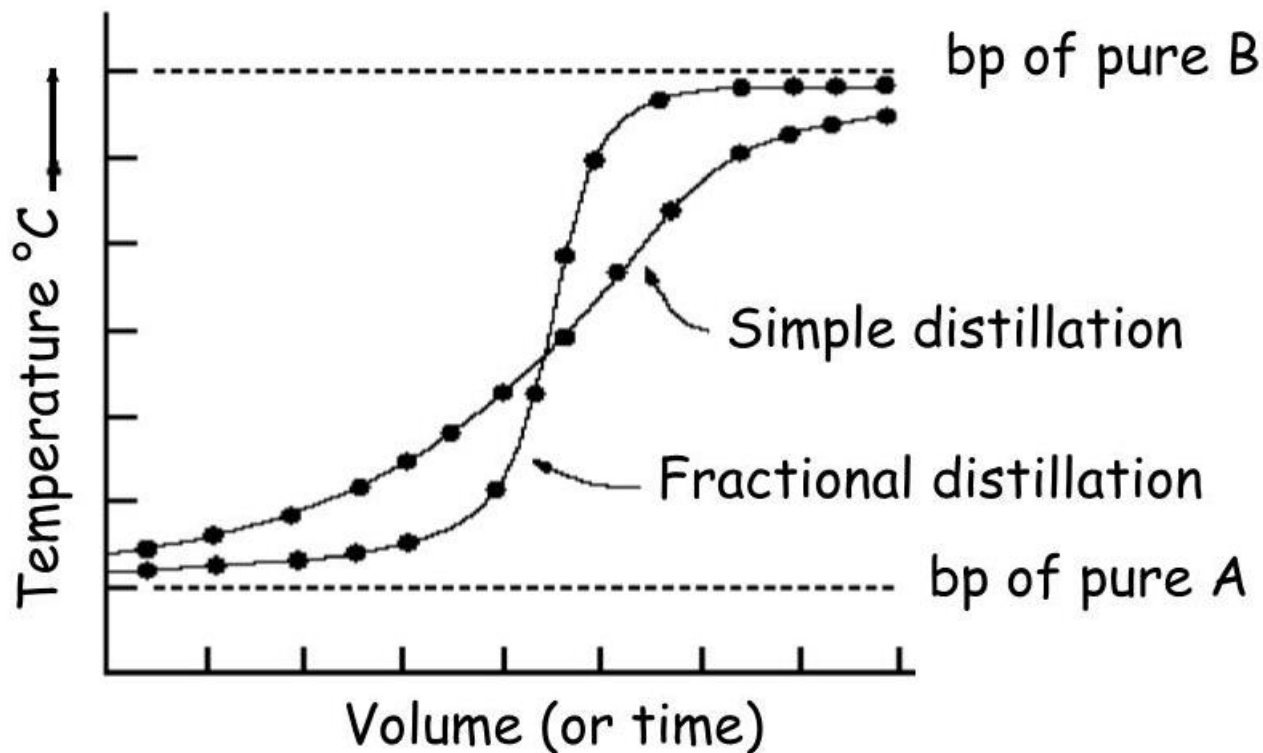


Fractional Distillation



Not in Atkins

Simple vs. Fractional



(Image: <http://www.chem.ucla.edu/~bacher/Specialtopics/Distillation.html>)

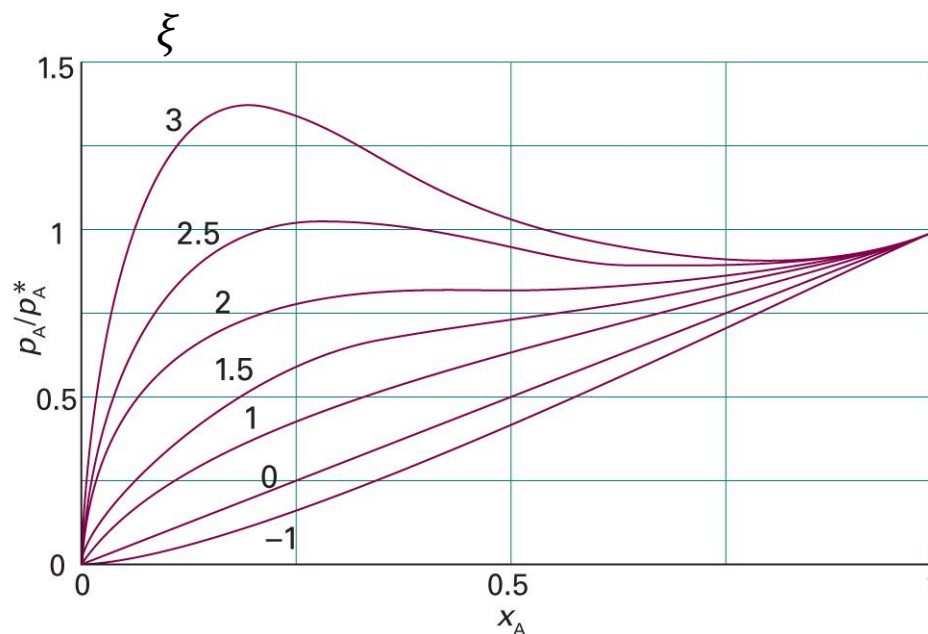
Real Solutions

$$\xi = \frac{z}{k_B T} \left(w_{AB} - \frac{w_{AA} + w_{BB}}{2} \right) \neq 0 \text{ (BW model)}$$

- $\xi > 0$: AB interactions are weaker
 - $\xi < 0$: AB interactions are stronger
1. Change in liquid-gas equilibrium: azeotrope
 2. Change in liquid-liquid equilibrium: phase separation

Vapor Pressure of BW Model

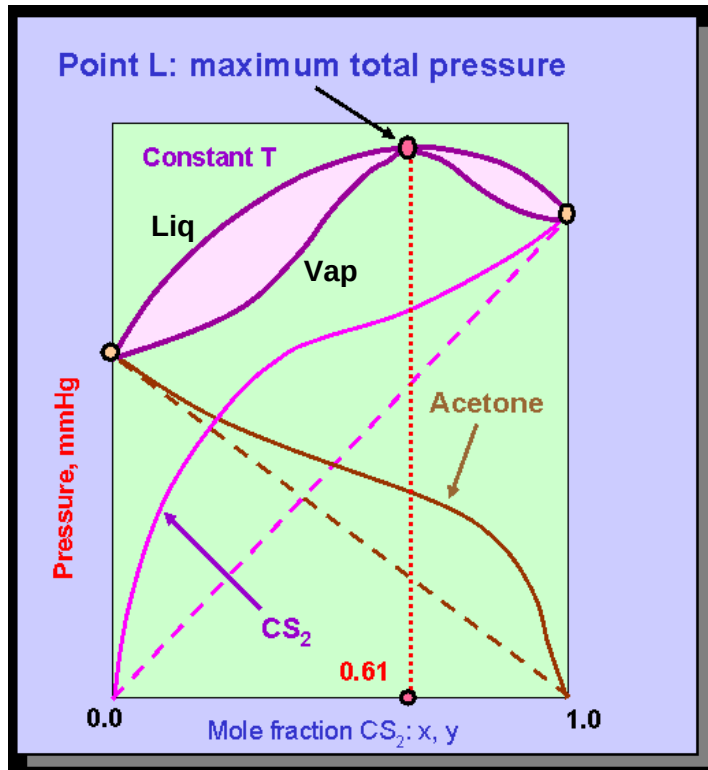
$$p_A/p_A^* = x_A e^{\xi(1-x_A)^2}$$



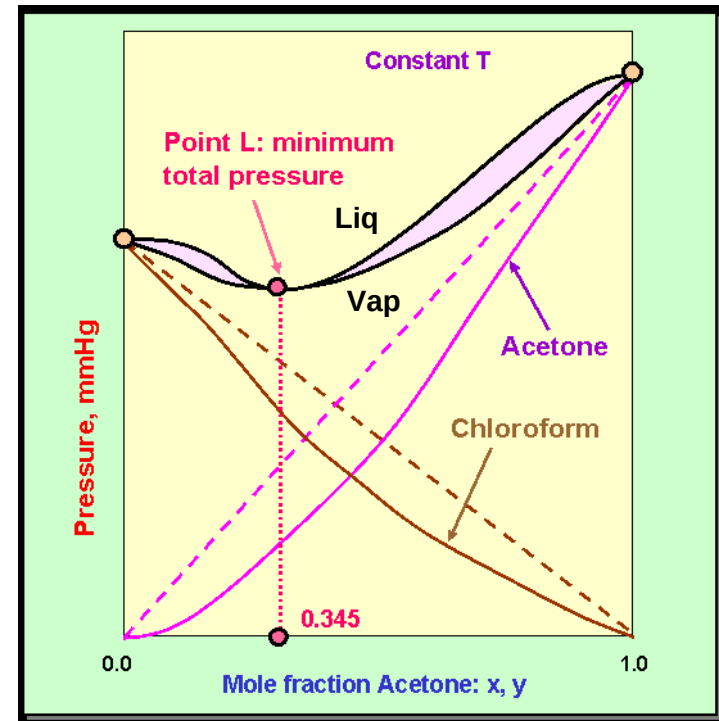
(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5F.2)

Non-Monotonic Behavior of p_{tot}

$$\xi > 0$$



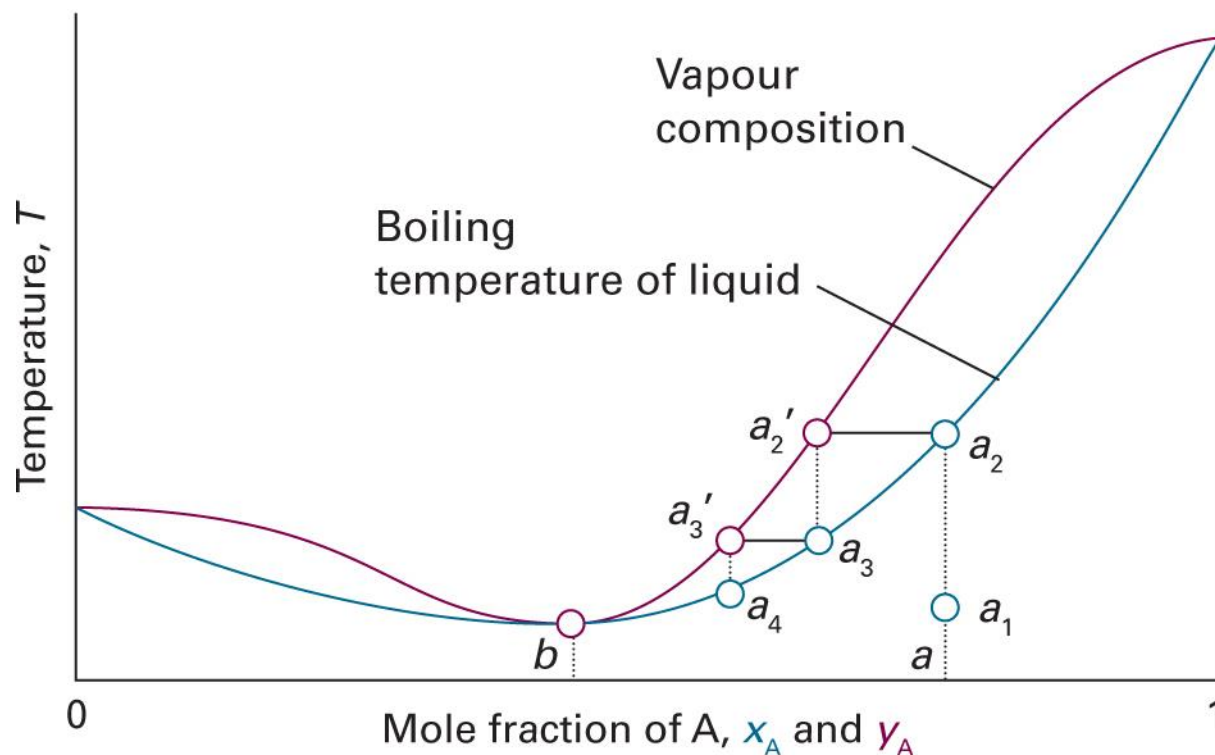
$$\xi < 0$$



(Image: http://www.separationprocesses.com/Distillation/DT_Ch01f.htm)

Low-Boiling Azeotrope

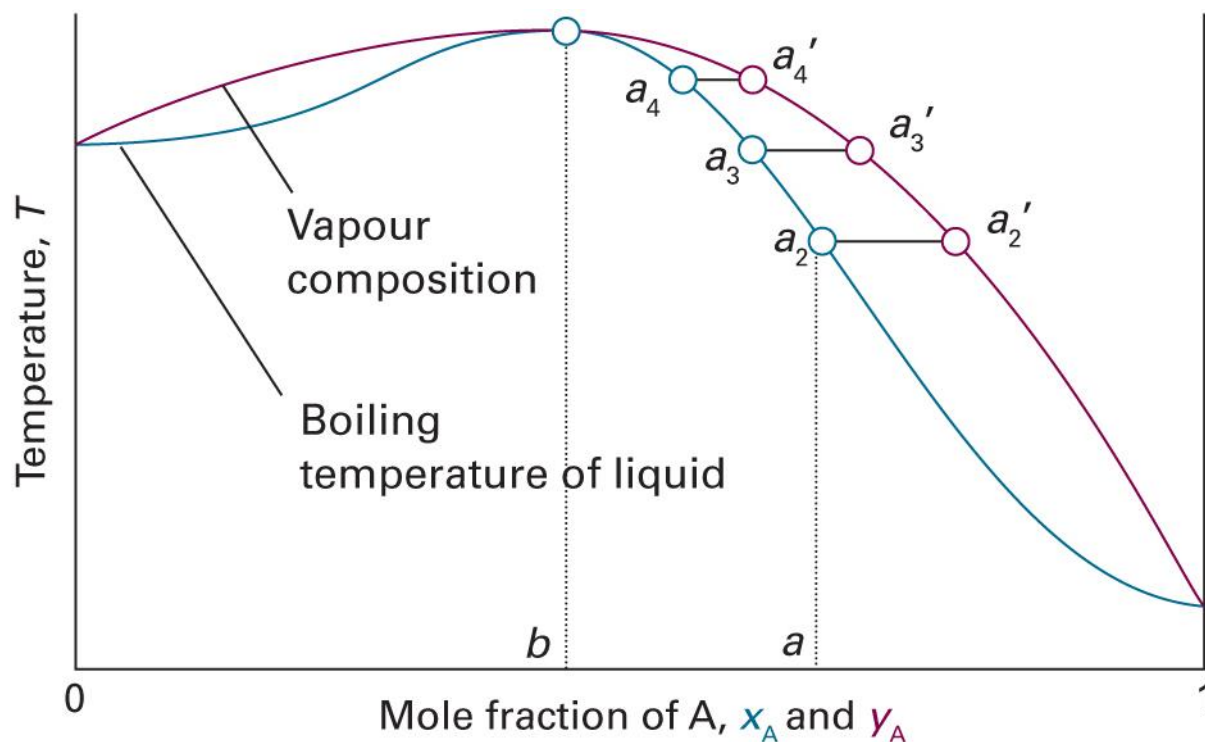
$$(\xi > 0)$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 12th ed., Figure 5C.12)

High-Boiling Azeotrope

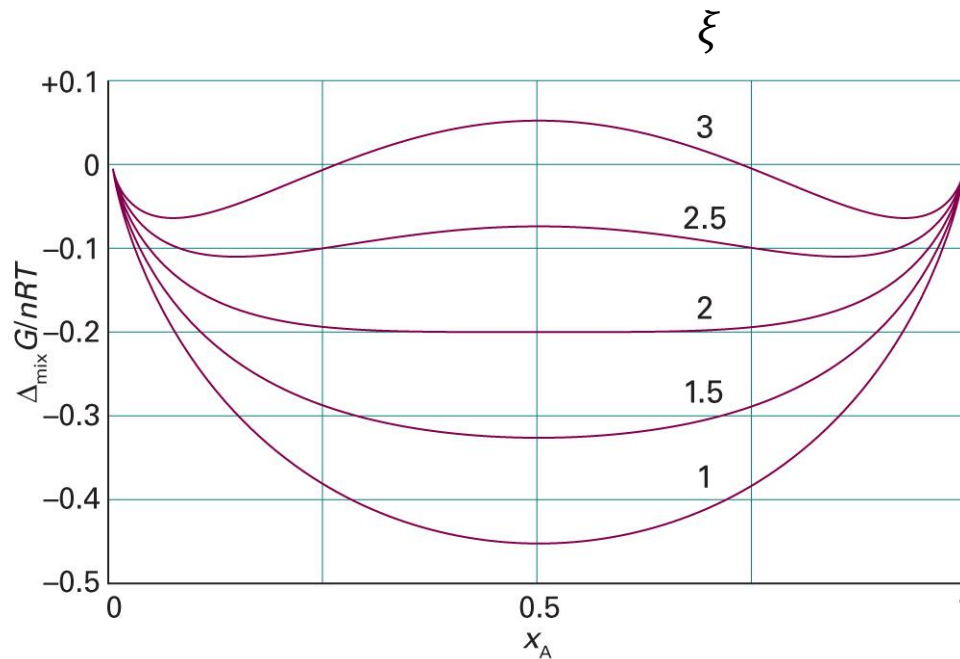
$$(\xi < 0)$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 12th ed., Figure 5C.11)

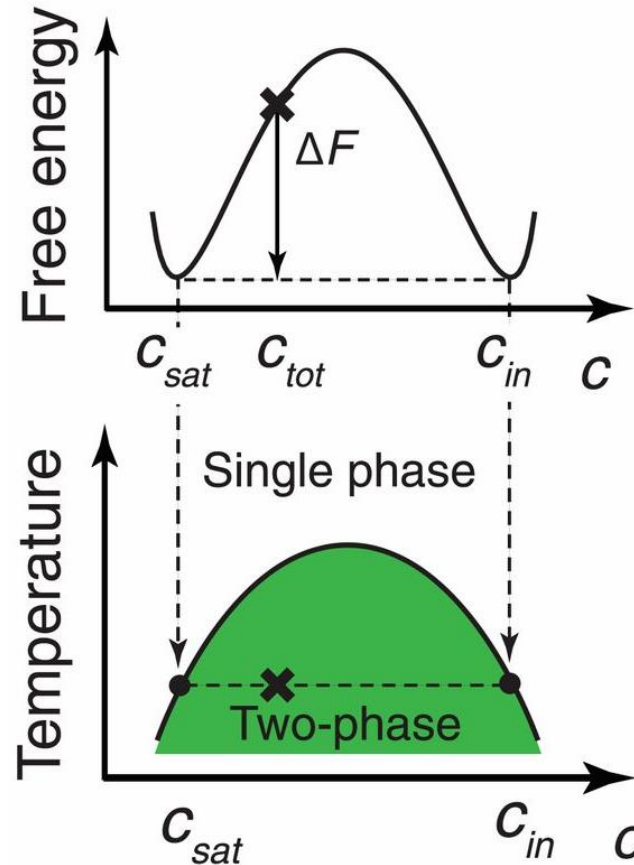
BW Model: Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5B.5)

What Happens at $\xi > 2$?



(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

Locations of Minima

$$\begin{aligned}\Delta_{\text{mix}}G &= nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B) \\ &= nRT\{x_A \ln x_A + (1 - x_A) \ln(1 - x_A) + \xi x_A(1 - x_A)\}\end{aligned}$$

The local minima can be found by using the derivative:

$$\begin{aligned}\left(\frac{\partial \Delta_{\text{mix}}G}{\partial x_A}\right)_{T,p} &= nRT\{\ln x_A + 1 - \ln(1 - x_A) - 1 + \xi(1 - 2x_A)\} \\ &= nRT\left\{\ln \frac{x_A}{1 - x_A} + \xi(1 - 2x_A)\right\} = 0\end{aligned}$$

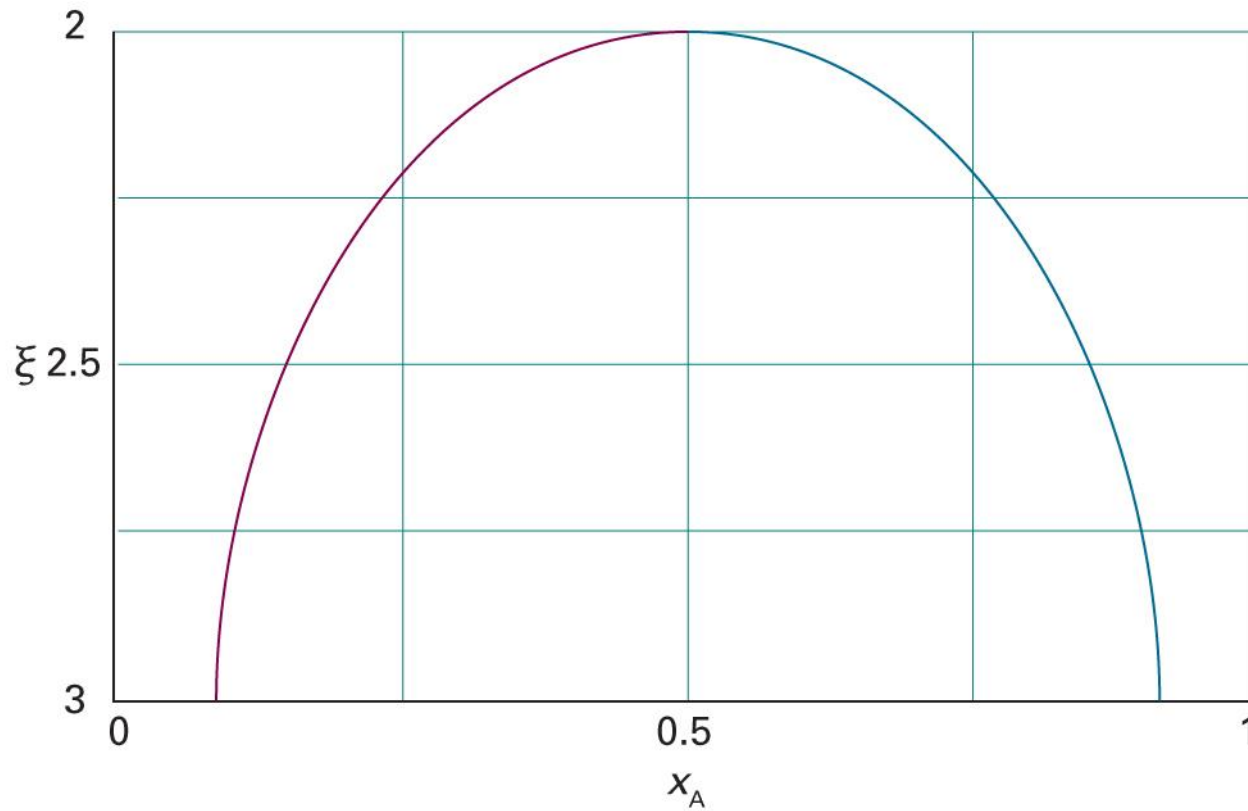
Locations of Minima

$$nRT \left\{ \ln \frac{x_A}{1 - x_A} + \xi(1 - 2x_A) \right\} = 0$$

The minima are located at x_A such that

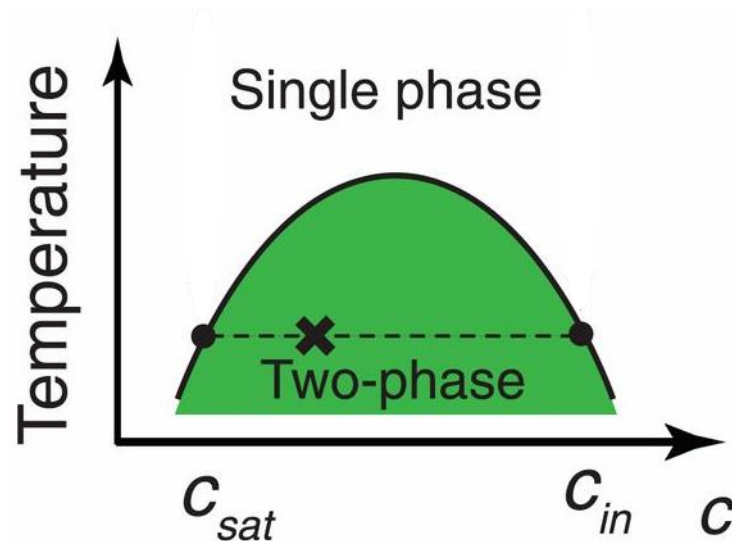
$$\ln \frac{x_A}{1 - x_A} = -\xi(1 - 2x_A)$$

Locations of Minima



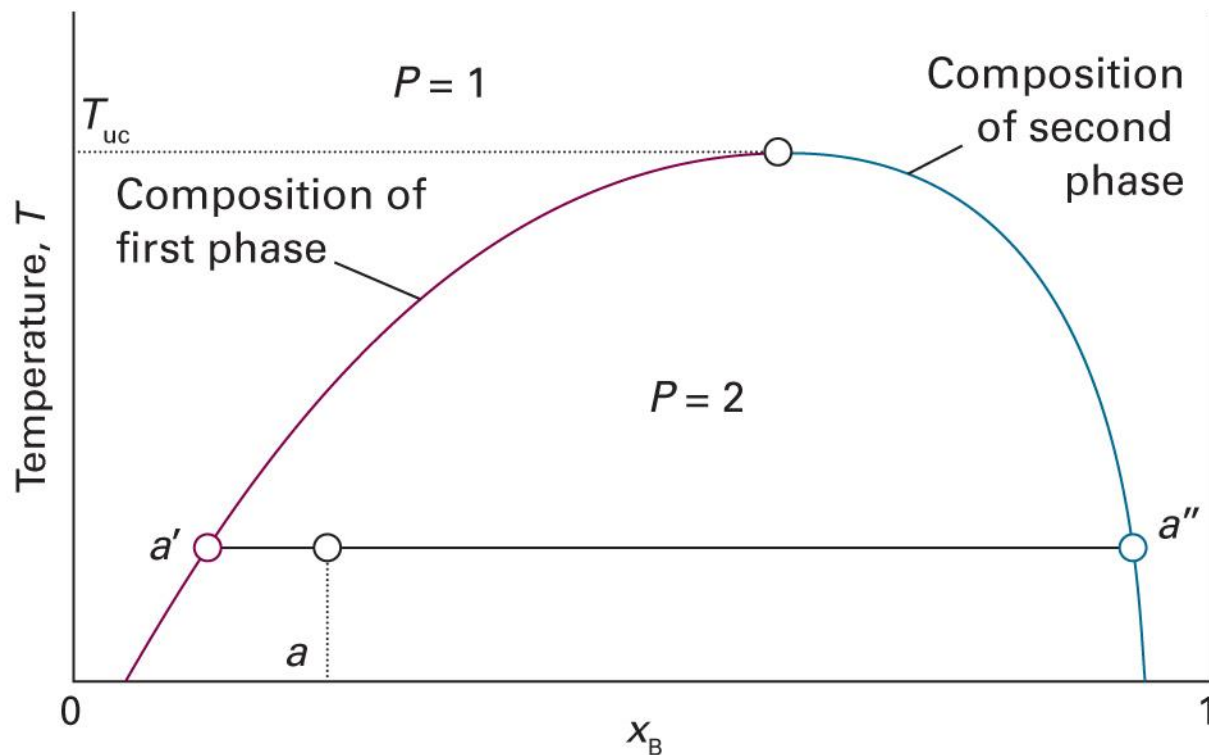
(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 12th ed., Figure 5C.18)

Phase Diagram



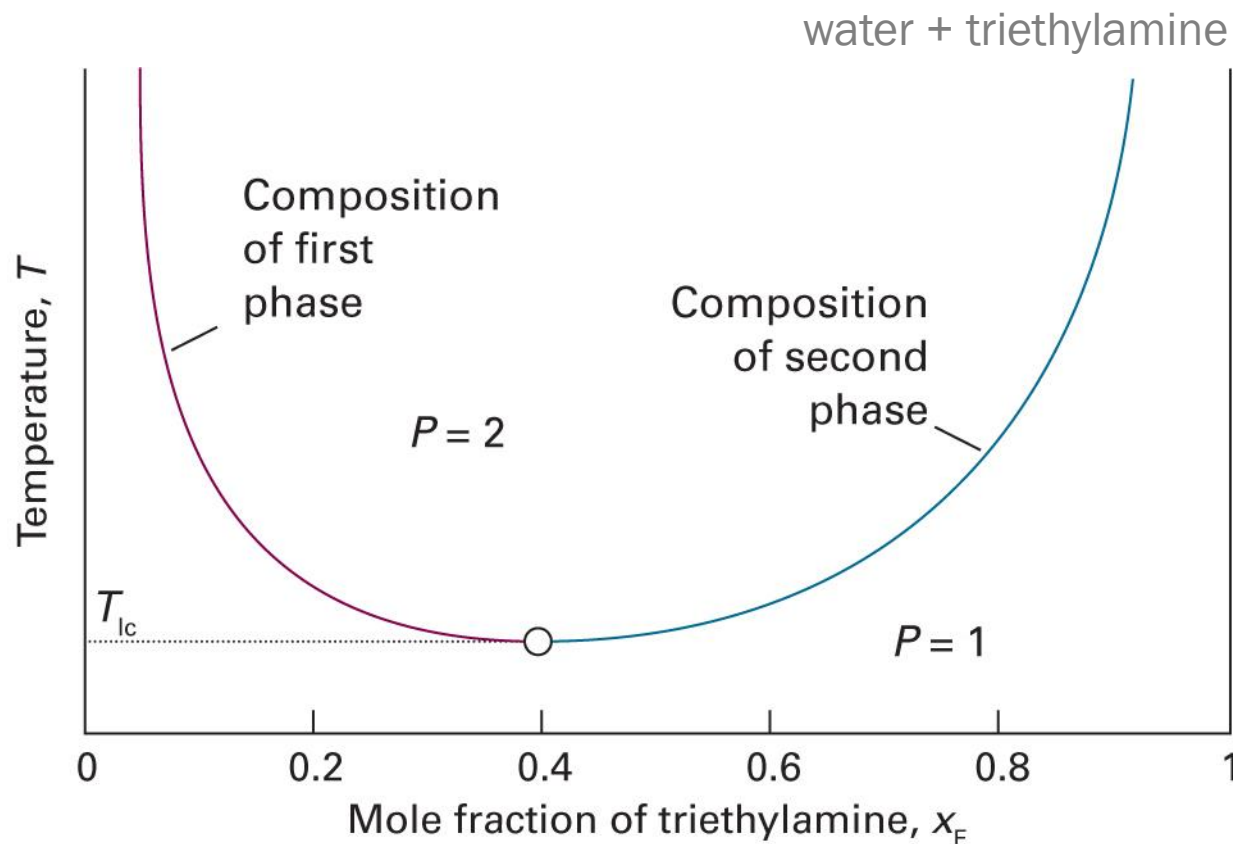
Lever rule applies in the two-phase region!

Upper Critical Solution Temperature



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5C.14)

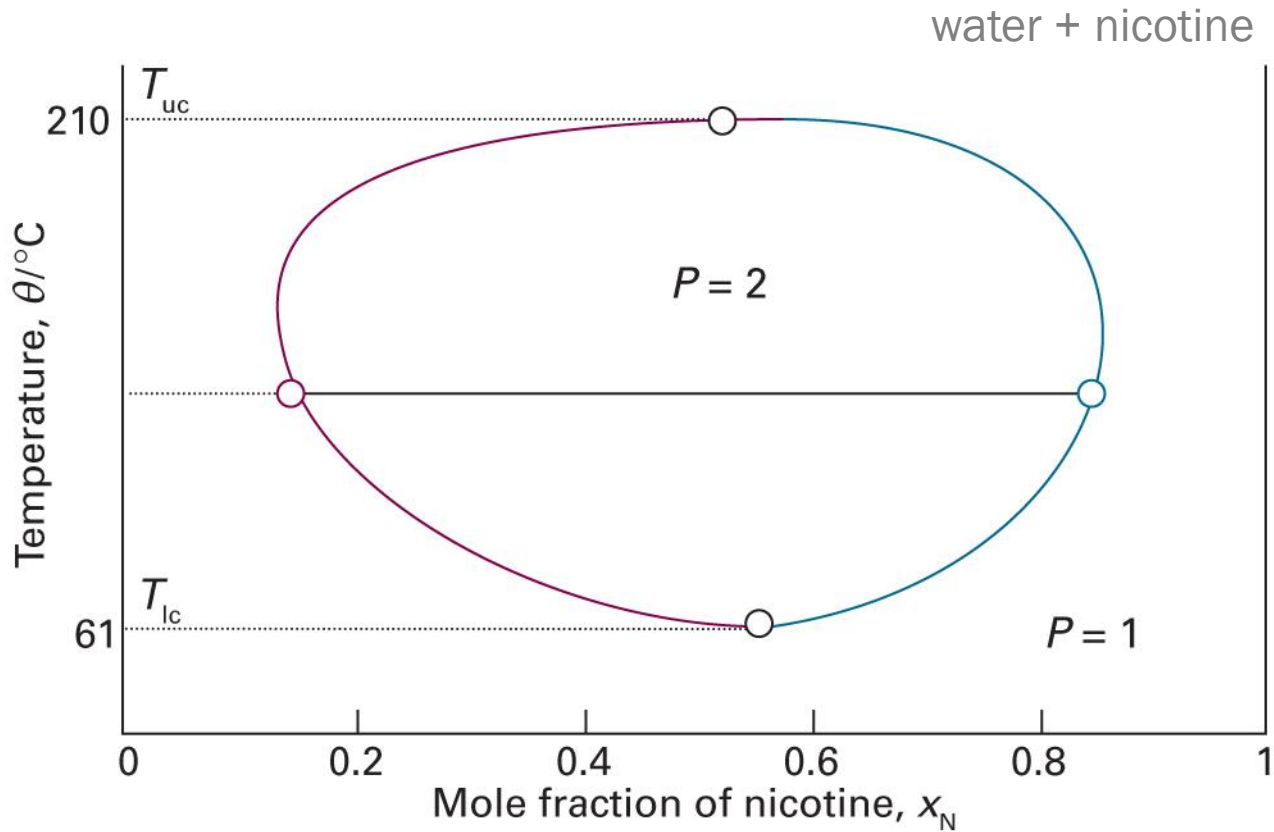
Lower Critical Solution Temperature



Two components form **weak complexes** at low T

(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 12th ed., Figure 5C.19)

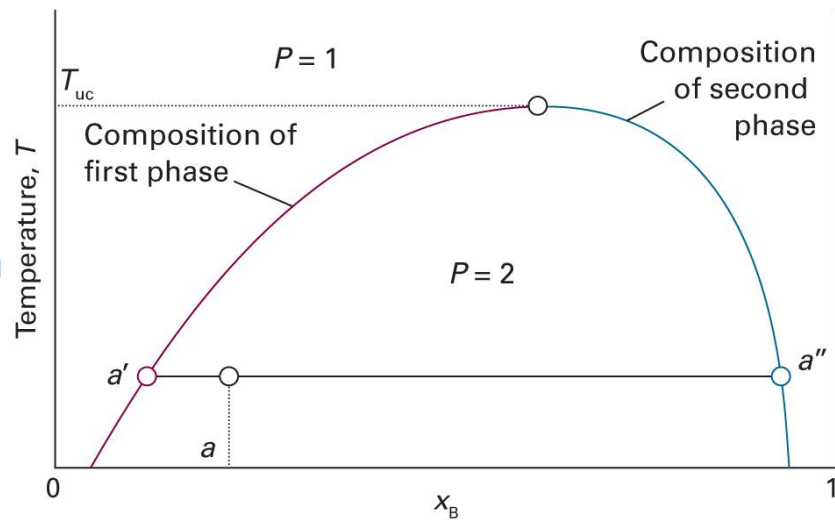
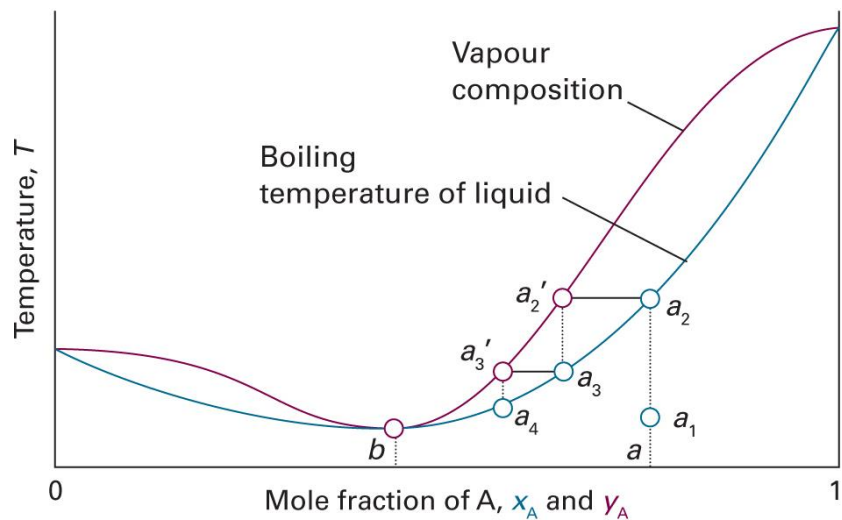
Both UCST and LCST



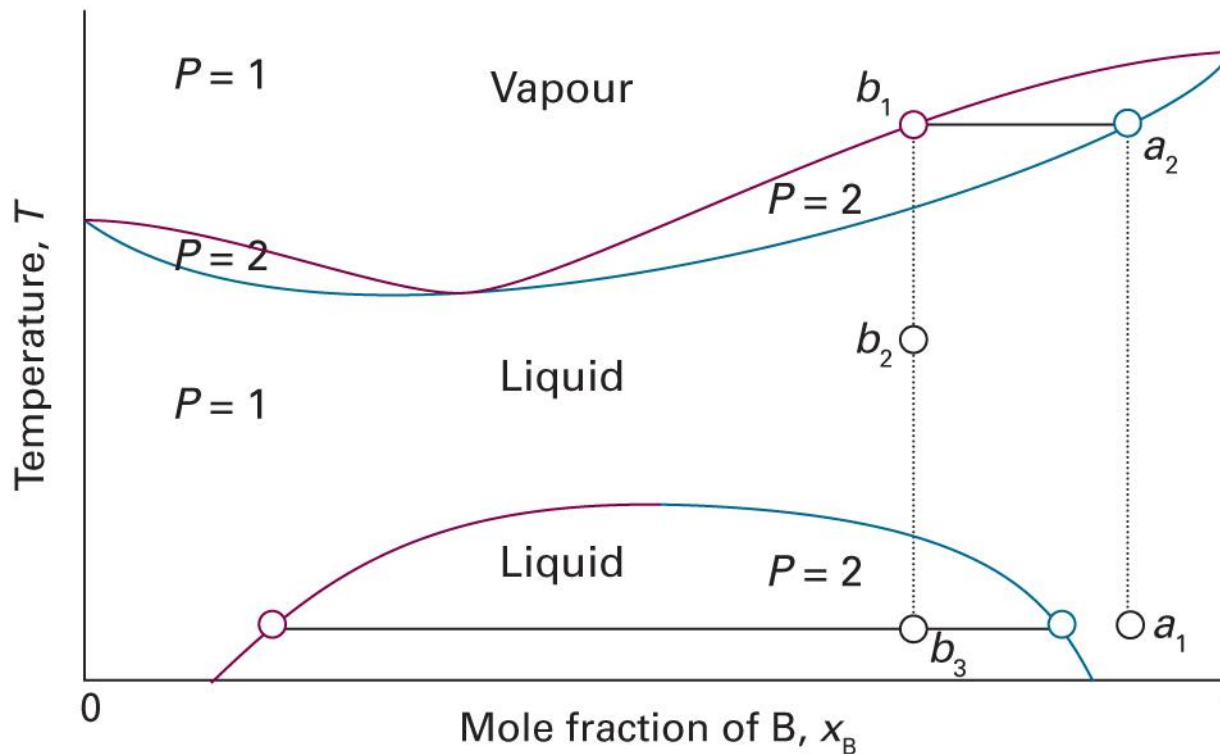
(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5C.20)

Azeotrope + Phase Separation

(when $\xi \gg 0$)

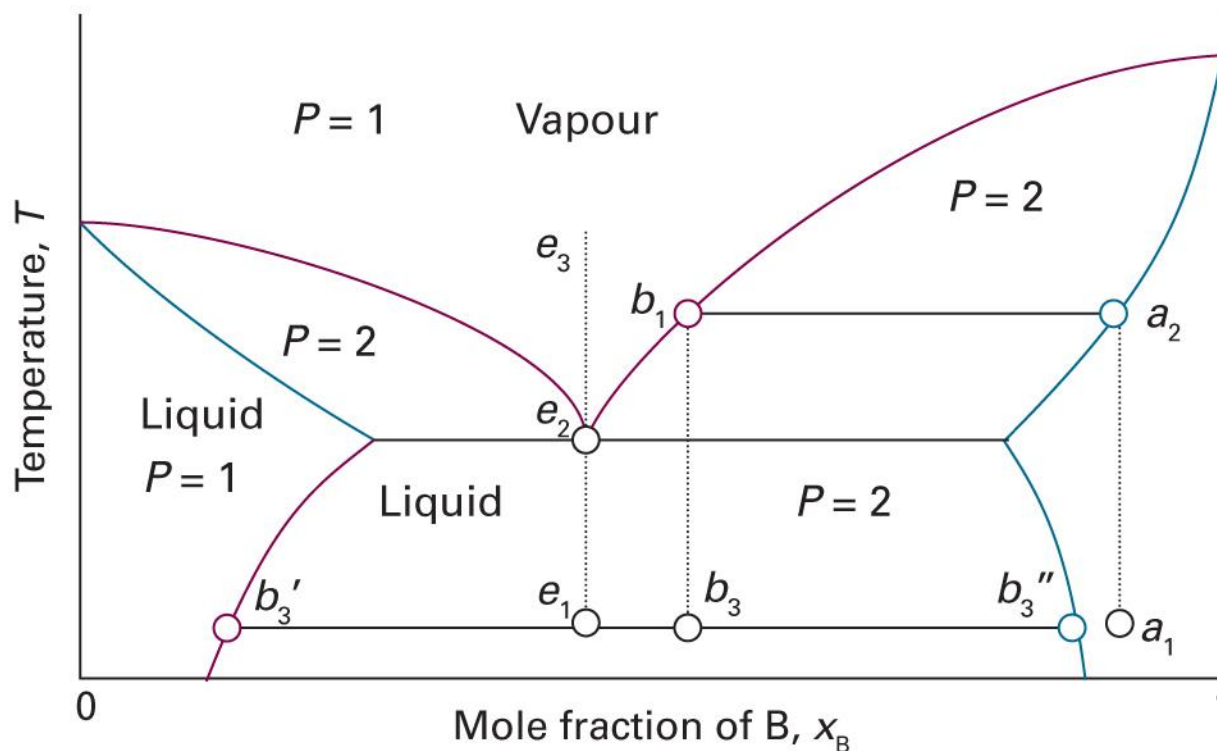


Case 1: Miscible before Boiling



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5C.21)

Case 2: Boiling before Miscible



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5C.22)